A RELATIVE EFFECTIVE CARTIER DIVISOR OVER A LOCALLY FINITELY PRESENTED BASE SITS IN AN OPEN SUBSCHEME THAT IS FLAT OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 4081-4088 and proof span 4090-4134 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof names its approximating rings with the author's own indexing and writes the two identities exhibiting the model by base change; both are preserved.

Editorial note. 5 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The approximation of a flat and finitely presented algebra by a flat algebra of finite type over a finitely generated ring, cited from the algebra chapter. Cited at proof lines 4098-4101. Produces the Noetherian model over which the quotient by the local equation stays flat.
- Surjectivity of the base change map on the kernel of multiplication by an element, cited from the more-algebra chapter. Cited at proof lines 4105-4107. Shows that the finitely generated kernel of multiplication by the equation over the model is killed after enlarging the base ring, since the kernel over the original ring vanishes.
- Flatness descends along the base change defining the model, cited from the morphisms chapter. Cited at proof lines 4116-4119. Transfers the flatness statement obtained for the model back to the original morphism.
- A relative effective Cartier divisor is flat at every point of its support, and the flat locus of a finite type morphism of Noetherian schemes is open, cited from the same chapter and the algebra chapter. Cited at proof lines 4123-4128. Gives pointwise flatness in the reduced situation and turns it into the required open neighbourhood.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02JO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3343–3343. Same-chapter reference used in the retained author text at source line(s) 4106, 4234, 5430. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original theorem 00RC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 998–998. Same-chapter reference used in the retained author text at source line(s) 4132, 4252, 5481. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 062U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7808–7808. Same-chapter reference used in the retained author text at source line(s) 4130, 5324. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 063Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7839–7839. Same-chapter reference used in the retained author text at source line(s) 4113, 5441. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01U8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2401–2401. Same-chapter reference used in the retained author text at source line(s) 4126, 5474. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $D \subset X$ be a relative effective Cartier divisor. If f is locally of finite presentation, then there exists an open subscheme $U \subset X$ such that $D \subset U$ and such that $f|_U : U \to S$ is flat.*

Proof. Pick $x \in D$. It suffices to find an open neighbourhood $U \subset X$ of x such that $f|_U$ is flat. Hence the lemma reduces to the case that $X = \mathrm{Spec}(B)$ and $S = \mathrm{Spec}(A)$ are affine and that D is given by a nonzerodivisor $h \in B$. By assumption B is a finitely presented A-algebra and B/hB is a flat A-algebra. We are going to use absolute Noetherian approximation.

Write $B = A[x_1, \ldots, x_n]/(g_1, \ldots, g_m)$. Assume h is the image of $h' \in A[x_1, \ldots, x_n]$. Choose a finite type $\mathbf{Z}$-subalgebra $A_0 \subset A$ such that all the coefficients of the polynomials $h', g_1, \ldots, g_m$ are in A_0. Then we can set $B_0 = A_0[x_1, \ldots, x_n]/(g_1, \ldots, g_m)$ and h_0 the image of h' in B_0. Then $B = B_0 \otimes_{A_0} A$ and $B/hB = B_0/h_0B_0 \otimes_{A_0} A$. By Algebra, Lemma 02JO we may, after enlarging A_0, assume that B_0/h_0B_0 is flat over A_0. Let $K_0 = \mathrm{Ker}(h_0 : B_0 \to B_0)$. As B_0 is of finite type over $\mathbf{Z}$ we see that K_0 is a finitely generated ideal. Let $A_1 \subset A$ be a finite type $\mathbf{Z}$-subalgebra containing A_0 and denote B_1, h_1, K_1 the corresponding objects over A_1. By More on Algebra, Lemma 063Q the map $K_0 \otimes_{A_0} A_1 \to K_1$ is surjective. On the other hand, the kernel of $h : B \to B$ is zero by assumption. Hence every element of K_0 maps to zero in K_1 for sufficiently large subrings $A_1 \subset A$. Since K_0 is finitely generated, we conclude that $K_1 = 0$ for a suitable choice of A_1.

Set $f_1 : X_1 \to S_1$ equal to Spec of the ring map $A_1 \to B_1$. Set $D_1 = \mathrm{Spec}(B_1/h_1B_1)$. Since $B = B_1 \otimes_{A_1} A$, i.e., $X = X_1 \times_{S_1} S$, it now suffices to prove the lemma for $X_1 \to S_1$ and the relative effective Cartier divisor D_1, see Morphisms, Lemma 01U8. Hence we have reduced to the case where A is a Noetherian ring. In this case we know that the ring map $A \to B$ is flat at every prime $\mathfrak{q}$ of $V(h)$ by Lemma 062U. Combined with the fact that the flat locus is open in this case, see Algebra, Theorem 00RC we win. □